



Apple doesn't keep doctor away?

An apple actually carries about 100 million bacteria. So if you eat apples whole, good luck!
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Science-approved petting method

The best way to pet a cat is by allowing it have as much choice and control as possible.
<https://digit.in/aug19-14>

makes them move away or towards light, and photophobia which prevents them from passing a boundary between light and dark areas. Researchers in 2002 isolated proteins that were associated with the behavior.

Now, if the same algal proteins could be incorporated into the nerve cells of mammals, then it would be possible to micromanage the nerve activity within living brains. All the researchers had to do was turn on the light, and the neurons would fire. The method was more accurate than electrical stimulation of brain tissue, and faster than the use of drugs. Researchers accomplished this by taking the genes that encode the protein from the algae, and put them in a virus. The virus is then injected into the brain of a mouse, where it infects neurons. The membranes of the neural cells in the mice then express the algal protein.

A fibre optic cable can then deliver the light stimulus to the brains of the mice. The researchers can target specific neurons in living mice using this method. The result is that researchers get to directly control the behavior of mice. A switch triggers a blue light in the brain of a mouse, and it starts moving in circles. This process is known as "optogenetics". Researchers have used the proteins within other organisms as well. In flies, the method was used to trigger the escape response, and a nematode was made to move or stay still. Different types of proteins can also be used to make sure that the neurons do not get fired, or stop them from firing if necessary. Essentially, scientists can very precisely control the behavior of organisms. The approach has already been used to provide a normal steady gait to a rat with Parkinson's disease. A similar strategy can potentially be used to treat a range of mental illnesses, including problems with memory, depression and addiction. Nature called it the method of the year for 2010.

Now, the method had its own restrictions too. Light tends to scatter within the brain matter, and can only reliably reach about half a millimeter

into the tissue. Also, sticking a fibre optic cable into the brain directly damages it. Researchers from the University of Riken in Japan have come up with a work around, that uses infrared light instead of light in the visible range. Nanoparticles introduced into the brain tissue at the target region, upconvert the infrared light to the visible range to the area that requires optical stimulation. Using this approach, researchers were able to reach 4.5 millimeters into the brain tissue. The approach was used in mice to induce memory recall and modify its fear response. The nanoparticles used for upconversion were made of heavy metals, which are toxic to mammalian tissue. The researchers coated the particles with silica to make them biocompatible. Also, infrared light can cook the delicate brain tissue, but the researchers discovered that the blood vessels act like heat sinks, that move away the excess heat

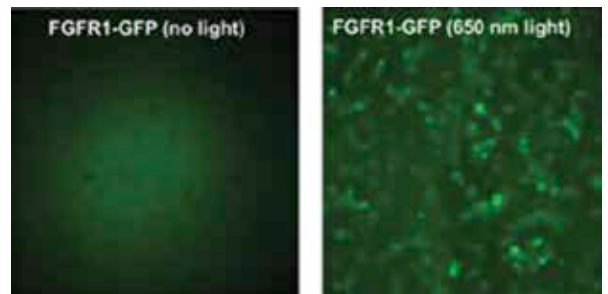
away from the brain. The researchers want to use this method to probe the brain cells of infant mice, and study the brain stem in much greater detail. After plenty of testing, we could have similar treatments in the future that can be used on humans to treat chronic pain, memory loss, restore eyesight or even reverse paralysis.

TOGGLING GENES

For over 20 years, researchers have been using laser light to control cellular interactions, to correct any problems in intercellular communication. So far there have been no methods developed to directly fix any problems in the genetic code. A new approach addresses this problem by combining optogenetics with nanotechnology.

Researchers from the University at Buffalo first grew brain tissue in the lab using stem cells. This tissue was embedded with implants, tiny wireless devices with lasers and antennas. The

wireless "switches" were triggered with various kinds of laser lights, a blue laser, a red laser and a far red laser. The interactions of the laser triggers and the implants allowed the researchers to activate and deactivate a particular gene, Fibroblast Growth Factor Receptor 1 or FGFR1 at will. FGFR1 controls about 4,500 other genes, or about one fifth of the human genome. It plays an important role in the development of an individual, right from the embryo to the adult. Co author of the study, Michal K. Stachowiak explains, "In some respects, it's like a boss gene. By controlling FGFR1, one can theoretically prevent widespread



On the left is the FGFR1 gene in its natural state. On the right is the gene exposed to laser light, which can activate and deactivate it.

Credit: University at Buffalo

gene dysregulations in schizophrenia or in breast cancer and also in other types of cancer."

The researchers have essentially used wireless technology to hack a gene. The approach can open up new treatments for treating cancer and mental health disorders, or even prevent them from occurring at all in the first place. The emerging field of direct manipulation of genes using light is known as "optogenomics". Josep M. Jornet, another co-author of the study says, "The potential of optogenomic interfaces is enormous. It could drastically reduce the need for medicinal drugs and other therapies for certain illnesses. It could also change how humans interact with machines."

The researchers are planning to introduce nano detectors onto the implants in the future. The next steps for the research is further testing on more lab grown brain tissue, as well as cancerous tissue. 